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A Co-distributive Account of Weak Reciprocity

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Abstract: On a weak reciprocity (WR) reading of *the children kicked each other*, each child kicks someone else and is kicked by someone else, but need not kick every other child or be kicked by every other child. This paper develops a co-distributive account of WR by extending cover-based analyses of non-maximality. For each participant, the reciprocal identifies all the other antecedent members, but context can permit some of them to be excluded from distribution. A cover determines which partners enter distribution. Allowing this cover to vary with the participant yields co-distributivity. Skolemized covers formalize this dependence while the distributive operators retain universal force. A worked derivation yields WR under cover choices that ensure every member participates in both relational roles. The analysis extends to possessive reciprocals and explains how subject *all* can require complete antecedent coverage while allowing non-maximal distribution over partners. These results provide a compositional route to WR without a dedicated cumulative operator. Further work must explain how predicate properties and conversational goals constrain cover selection, ruling out unacceptable interpretations and accounting for speakers' preferences among the available readings.

Keywords: reciprocity; co-distributivity; cumulativity; Skolemized covers; non-maximality; homogeneity

1. Understanding weak reciprocity

Reciprocal expressions such as *each other* give rise to readings of different stre ngths. This paper focuses on weak reciprocity (WR), illustrated by the following e xample, adapted from Langendoen [1].

(1) *They scratched each other's backs.*

On a WR reading, each person scratched someone else's back and had their own back scratched by someone else. These partners need not be the same: the s entence can describe people standing in a circle, each scratching the next person's back. WR thus requires everyone to participate in both roles, without requiring ev ery possible pair.

Compare the strong reciprocal (SR) reading of (2).

(2) *The three men knew each other.*

On this reading, each man knew both of the others. SR requires the relation to hold between *every* pair of distinct members and therefore entails WR for plur al antecedents. The central question of this paper is how reciprocal sentences deri ve the weaker requirement of WR [1,2].

One influential approach derives the WR reading of (1) using a cumulative o perator [3,4]. We propose instead that WR can arise through non-maximal distribu tion, extending Brisson's [5] cover-based account of non-maximality. The reciprocal still identifies *all* the other members of the antecedent group, but the predicate n eed not apply to every one of them for each participant. A contextually supplied cover determines which partners are included in distribution.

Our proposal allows this cover to depend on the participant. In the circular backscratching situation, for example, each person's selected partner may be the p erson immediately ahead. Requiring each person to scratch that partner's back doe s not require every possible pair to interact. This dependence yields a co-distributi ve interpretation. It draws on Winter's [6] analysis of dependent definite plurals, while locating the dependence in the cover rather than in the definite description itself. We formalize this dependence using Skolemized covers, extending Schwarzs child's [7] cover framework.

To preserve WR, each participant must have at least one selected partner and must also be selected as someone else's partner. With these constraints, the resul ting readings satisfy cumulative coverage without requiring a dedicated cumulativ e operator in their composition.

Some reciprocal readings are weaker than WR [1,2]. For example, *the students followed each other* can describe a line in which one student follows nobody and a nother is followed by nobody. The reading therefore does not require every partic ipant to fill both roles. We suggest that such readings can arise through further n on-maximal weakening of the participation requirements. The main analysis develo ped here focuses on WR. Figure 1 illustrates the distinction between SR, WR, and a directed line that fails WR.

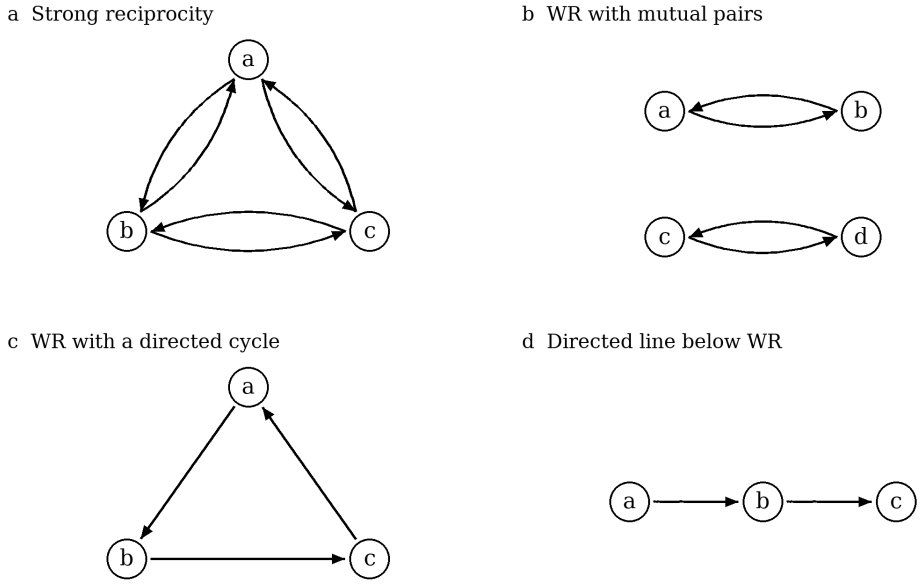


Figure 1. Reciprocal relation patterns. An arrow from x to y represents R(x, y). Panel (a) satisfies SR. Panels (b) and (c) satisfy WR but not SR. Panel (d) is a directed line whose endpoints participate in only one role, so it fails WR.

Section 2 introduces cumulativity and co-distributivity and reviews the releva nt literature. Section 3 reconstructs Beck's cumulative analysis and examines posse ssive reciprocals and contrasts involving *all*. Section 4 develops the dependent-cov er analysis, presents the derivations, and discusses contextual selection and overge neration. Section 5 summarizes the contribution and remaining questions.

2. Cumulativity and dependent distribution

2.1. Cumulative coverage and cumulative composition

Cumulative readings allow two pluralities to be related without requiring eve ry member of one to be related to every member of the other [8].
Consider the classic example in (3), from Scha [9].

(3) *600 Dutch firms use 5000 American computers.*

On a cumulative reading, each of the 600 firms uses at least one of the 5000 computers, and each of those computers is used by at least one of the firms. Diff erent firms may use different numbers of computers, and several firms may share a computer. The sentence does not require every firm to use every computer.

Let D be the domain of atomic[1] individuals, let A and B be nonempty finite subsets of D, and let $R \subseteq D \times D$. We define the cumulative coverage condition i n (4), adapted from Beck and Sauerland [12] (p. 351, Eq. 3).

(4) $\mathrm{CUMU}(R)(A, B) \Leftrightarrow$
$[\forall x \in D\ (x \in A \rightarrow \exists y \in D\ (y \in B \wedge R(x, y)))]$
$\wedge\ [\forall y \in D\ (y \in B \rightarrow \exists x \in D\ (x \in A \wedge R(x, y)))]$.

Condition (4) states what the cumulative reading of (3) requires, but leaves o pen how those requirements arise. This matters for the reciprocal analysis: cumula tive coverage does not by itself establish the need for a dedicated operator. In op erator-based analyses, $**$ closes a binary relation under coordinatewise sum format ion [3]. Beck and Sauerland [12] allow $**$ to apply to non-lexical relations constru cted by QR and predicate abstraction. Champollion [13] retains $**$ in an eventless analysis of cumulative readings of every.

Event-based analyses connect cumulative coverage to plural events and their participants. In Landman's [14] analysis, plural thematic roles map an event sum t o the summed role values of its atomic subevents. Chatain [15] develops a trivale nt event semantics with separate homogeneity effects for agent and theme roles, a llowing their exhaustive participation inferences to disappear independently under negation.

Schmitt [16] locates cumulativity in function–argument composition rather tha n in an operator applied to a relation. Atomic function and argument parts are p aired so that every part is used at least once, yielding possible sums of the result ing values. Cumulative structure thus develops through successive composition ste ps. Schmitt [17] extends this plural projection approach to attitude contexts, where pluralities of individual concepts project to pluralities of propositions.

Chatain's dissertation [18] (Chapter 2) develops a separate, eventless analysis based on weak meanings and strengthening. Verbal predicates have weak existenti al meanings; recursive exhaustification derives exhaustive participation inferences. For cumulative readings of every, subplurality alternatives and the every/some sca le yield these inferences as distributive implicatures. This account retains standard quantifier meanings and relatively standard composition, without requiring $**$ or a cumulative composition rule.

These alternatives motivate the distinction emphasized by Haslinger et al. [19]: whether cumulativity extends beyond lexical predicates and whether the gramma r contains a dedicated cumulation operator. Non-lexical cumulative readings need

[1] What counts as atomic is a matter of debate in physics and philosophy. Here we adopt a simple, intuitive understanding of atomic individuals as single units that cannot be further decomposed under the individual-part relation relevant to our analysis (cf. Link [10]). This intuition of unity is also related to Moltmann's [11] notion of an integrated whole, which need not lack internal parts. Atomicity in this sense does not imply physical indivisibility.

not involve such an operator, as Schmitt's plural projection illustrates. The locality argument in Section 3.2 therefore targets the QR operations needed to construct the input to ** in Beck's [4] reciprocal analysis. Its implications for other architectures depend on whether they require the same extraction step.

2.2. Dependencies between distributive domains

Co-distributivity describes readings in which participants relate to their respective counterparts without every cross-pair being required to satisfy the relation. Following Winter [6] (Section 2.2) and Minor [20] (Section 1), the term is used descriptively, leaving its compositional implementation open. Winter's dependency analysis provides the starting point for the present proposal. Consider his example in (5).

(5) *The soldiers hit the targets.*

On Winter's dependency reading, each soldier is associated with an assigned set of targets. Subject distribution binds a variable that supplies the argument of a contextually salient function from soldiers to their targets. The object definite consequently denotes the targets associated with the soldier currently being evaluated. Distribution over that local set requires the soldier to hit every target in it. Unary distribution and dependent definite reference thus derive the co-distributive reading without a polyadic distributive operator [6] (Section 3.1).

Winter [6] (Section 3.5) also observes that a morphologically plural dependent can have a singleton value. This connects his discussion to dependent-plural readings such as (6), from Minor [20].

(6) *All the girls were wearing hats.*

Each girl may wear a single hat, while more than one hat is involved overall. The local correspondence therefore need not itself involve a plurality. Zweig [21] accounts for overall multiplicity within a cumulative event semantics, combining number-neutral bare plurals with a multiplicity implicature computed before existential event closure. The licensing of dependent plurals by *all* and *most* remains open in that account (Section 8.6, n. 38). Minor [20] instead distinguishes weak distribution over a shared information state from strong distribution over separate states for the distributed individuals. In interaction with multiplicity implicatures, this distinction allows plurality to be required across participants or separately for each participant, underpinning his account of the *all/every* contrast (Sections 5.4, 6.2–6.4).

The reciprocal analysis adapts Winter's dependency strategy while locating the dependence in a cover rather than in nominal reference. For each antecedent member, the reciprocal still denotes all the other members. A participant-dependent cover determines which of those members enter distribution: the predicate must hold of every selected partner, while non-maximality permits other pairs to be omitted. Strict WR follows under cover choices that provide a nonempty partner set for every participant and include everyone as somebody else's partner. Section 4 formalizes this extension using Skolemized covers and examines the unresolved problem of contextual cover selection.

3. Challenges for a cumulative operator analysis

3.1. Reciprocal semantics and Beck's cumulative derivation

The dependency strategy outlined above offers a way to derive WR without a dedicated cumulative operator. To assess this alternative, we first establish a common analysis of reciprocal content and reconstruct Beck's [4] cumulative derivation. We identify both the operator's semantic contribution and the LF configuration required by the analysis, providing the basis for the two arguments that follow.

Consider Beck's [4] (p. 102) example:

(7) *The children like each other.*

Beck [4] (pp. 90–92) develops a group-denoting version of Heim, Lasnik, and May's [22,23] classic decompositional analysis of English *each other*. On this analysis, the reciprocal has two anaphoric arguments: a range argument identifying the antecedent plurality and a contrast argument identifying the current participant. It denotes the other members of the antecedent, while quantificational force is supplied by the mechanisms of plural predication.

Let D_e contain atomic and plural individuals, let $D_{at} \subseteq D_e$ be the atomic individuals, and let $\leq$ be the part-of relation. Let A denote the finite antecedent plurality, containing at least two atoms. For an atomic contrast individual x and a singular or plural range individual Y, define:

(8) $O(x, Y) = \bigoplus\{z \in D_{at}: z \leq Y \wedge z \neq x\}$.

Here $\bigoplus$ returns the sum of the individuals in a nonempty set; O(x, Y) is undefined when that set is empty. The reciprocal's nominal component can then be represented schematically as:

(9) $[\![[\text{the other } x_i \text{ of } Pro_j]]\!]^g = O(g(i), g(j))$.

The range pronoun Pro_j supplies A, while the contrast variable x_i is bound by an operator ranging over antecedent members. For an atomic $x \leq A$, O(x, A) therefore denotes the sum of all antecedent members other than x. Beck's general distinctness condition is non-overlap; with atomic participants, this reduces to non-identity.

This nominal denotation does not itself require the predicate to hold of *every* member of that sum. Universal quantification comes from distribution. For a property P of atomic individuals and an individual X, define:

(10) $DIST(X)(P) = 1 \Leftrightarrow \forall u \in D_{at}\ (u \leq X \rightarrow P(u))$.

If R is the atomic relation expressed by like, distribution over both A and the reciprocal's denotation yields SR:

(11) $DIST(A)(\lambda x.DIST(O(x, A))(\lambda y.R(x, y))) = 1$

$\Leftrightarrow \forall x \in D_{at}\ [x \leq A \rightarrow \forall y \in D_{at}\ ((y \leq A \wedge y \neq x) \rightarrow R(x, y))]$.

Each child must thus like every other child. The question is how the same reciprocal content can support WR instead. Our central concern is this mechanism of interpretation; the reciprocal denotation serves as a common working assumption.

Beck derives WR by raising both the subject and the reciprocal-internal range pronoun, then applying ** to the resulting relation [4] (pp. 101–105, Section 4.3.1). We assume a cover containing every antecedent member. The schematic LF in (12) reconstructs her (110a), with indices. The object description abbreviates the nominal structure interpreted in (8).

(12) $[Pro_3\ [[\text{the children}]_3$
$[**\ [1\ [2\ [t_1 \text{ like } [\text{the other } x_1 \text{ of } t_2]]]]]]]$

Abstractor 1 binds both the subject trace t_1 and the contrast variable x_1. Abstractor 2 binds the trace t_2 left by movement of the range pronoun Pro_3. Coindexation represents Pro's obligatory anaphoric connection to the antecedent; additional antecedent QR can make this connection explicit through variable binding [4] (p. 71, n. 2).

Using subject-first argument order, the relation constructed below ** is:

(13) $K = \lambda x.\lambda Y.R(x, O(x, Y))$.

The cover restricts the input pairs to atomic individuals. Applying the cumul ated relation first to the subject and then to the antecedent-denoting Pro gives:

(14) $**K(A)(g(3)) = **K(A)(A)$, where $g(3) = A$.

The operator $**$ closes the positive extension of K under coordinatewise sum formation. Its true plural pairs are constructed by summing true atomic pairs, wit h their first and second coordinates summed separately. Consequently, every atom ic member of each plural argument must occur in at least one contributing pair.

For atomic x and Y, O(x, Y) equals Y when $x \neq Y$. When $x = Y$, the descrip tion is empty and undefined. K therefore presupposes distinctness and asserts R(x, Y), rather than asserting their conjunction. Beck supplements pluralization with d efinedness conditions for partial functions [4] (pp. 103–104). Under the assumed c over, the positive truth conditions are:

(15) $**K(A)(A) = 1 \Leftrightarrow$

$[\forall x \in D_{at} (x \leq A \rightarrow \exists y \in D_{at} (y \leq A \wedge x \neq y \wedge R(x, y)))]$

$\wedge [\forall y \in D_{at} (y \leq A \rightarrow \exists x \in D_{at} (x \leq A \wedge x \neq y \wedge R(x, y)))].$

These are the strict WR conditions: every antecedent member participates in b oth relational roles, without every distinct pair being required. The inequalities in (15) identify the distinct atomic pairs available for cumulation; they do not turn d istinctness into an asserted component of K.

In this derivation, $**$ supplies two-way coverage, while the reciprocal's intern al description supplies anaphoric dependence and distinctness. Crucially, the input to $**$ is the non-lexical relation K, constructed by movement from within the reci procal. Beck independently motivates cumulation from complex nominals [4] (pp. 106–107, Section 4.3.3), and includes covers to explain exception effects [4] (Section 4.4). The following subsections examine specific questions about this architecture: whether the required movement remains available inside possessive reciprocals, an d the availability of weak reciprocal readings with subject *all.*

3.2. Syntax and LF in possessive reciprocals

The occurrence of *each other* in prenominal possessor position provides a local ity test for Beck's QR-based derivation of WR. In (16), *each other* occurs inside the object DP rather than as the verb's direct object.

(16) *The students graded each other's papers.*

Consider four students, each with one relevant paper, organized into two rev iew pairs. Each student grades only the partner's paper. Sentence (16) can describ e this situation: every student grades someone else's work and has their own wor k graded, although no grading occurs across the two pairs. The question is wheth er Beck's QR-based architecture can derive this reading when the reciprocal occup ies a possessor position.

Quantificational possessors can support binding outside the possessive DP. Ba rker [24] (pp. 107–108) analyzes them through adjunction to their containing DP, r ather than directly to the clause. This makes it necessary to distinguish a quantifi er that constitutes the possessor from one embedded within a larger possessor. Th e examples in (17) probe this distinction. Brackets identify the relevant constituent s, and coindexation marks the intended bound-variable reading.

(17) a. *[No president]$_i$'s biography discusses his$_i$ death.*

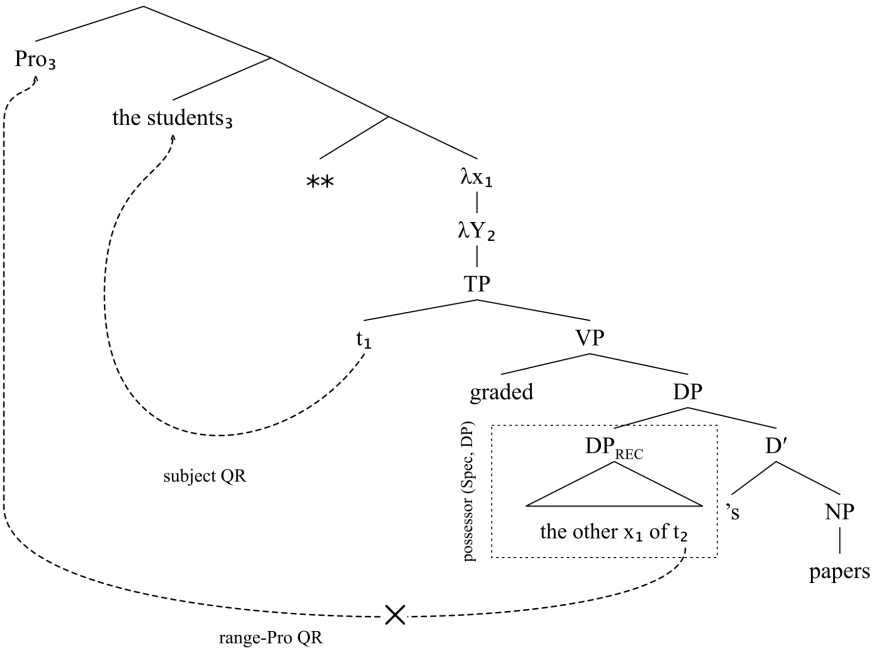
b. *# [Some biography about [no president]$_i$]'s final chapter discusses his$_i$ death.*

In (17a), *no president* constitutes the possessor and can bind *his*: there is no p resident x whose biography discusses x's death. In (17b), the same quantifier is e

mbedded within the larger possessor *some biography about no president*. The target r eading—that there is no president x such that the final chapter of some biograph y about x discusses x's death—is degraded. In these examples, the intended bindi ng is thus available when the quantifier constitutes the possessor, but degraded w hen it is embedded within one. Beck's reciprocal derivation requires movement fr om a similarly embedded position: the range pronoun occurs inside the reciprocal possessor. The relevant locality question is whether that pronoun can undergo the required QR.

Applying Beck's architecture to (16) raises both the subject and the reciprocal-internal range pronoun above **, as represented schematically in (18). The display extends the architecture introduced in Section 3.1 to the possessive example.

(18) [Pro_3 [[the students]$_3$
[** [1 [2 [t_1 graded [[the other x_1 of t_2]'s papers]]]]]]



Pro_3 is anaphorically linked to the antecedent *the students*. The abstraction ind ices 1 and 2 abbreviate λx_1 and λY_2, respectively: λx_1 binds the subject trace t_1 an d the reciprocal's contrast argument x_1, while λY_2 binds the range trace t_2. Crucia lly, t_2 is contained within the reciprocal possessor, which occupies Spec, DP. The derivation therefore requires extraction from inside the possessor.

Possessor binding does not independently establish clause-level possessor QR: Barker's [24] (pp. 107–108) analysis adjoins raised possessors to their containing D P rather than directly to the clause. Beck [4] (pp. 106–107, Section 4.3.3) independ ently motivates cumulation over relations constructed by QR from complex nomin als, but those examples do not by themselves establish the availability of possesso r-internal extraction in (18). If English blocks extraction of the internal range pron oun along the displayed path, this particular LF cannot generate the WR reading of (16). The locality concern is therefore conditional and specific to (18); it does n ot exclude cumulative analyses that construct the relevant relation differently.

The next subsection examines a further comparison involving *all*.

3.3. "All" and WR

Subject *all* provides a second point of comparison between WR and other plu ral readings. Its acceptability varies across reciprocal sentences and readings, but i t can be added naturally while preserving a WR reading, as illustrated in (19).

(19) a. *The four students graded each other's papers.*

b. *All four students graded each other's papers.*

Consider a peer-review activity involving four students, each with one paper. The students choose whose paper to grade, subject to two requirements: each gra des exactly one other student's paper, and each paper is graded by exactly one p eer. Figure 2 illustrates one possible outcome. Both sentences in (19) can describe this situation. Every student participates as a grader and as an author whose pap er is graded. Adding *all* therefore preserves WR without requiring every student t o grade every other student's paper.

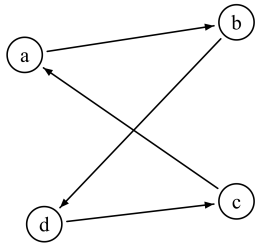


Figure 2. (All) the four students graded each other's papers.

Compare the weaker reading available with *follow*.

(20) a. *The three students followed each other.*

b. # *All three students followed each other.*

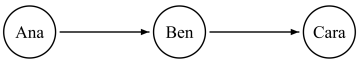


Figure 3. The three students followed each other.

Suppose Ana follows Ben and Ben follows Cara, with no other following rela tions. Sentence (20a) can describe this directed line. The situation fails strict WR: Cara follows nobody, and Ana is followed by nobody. (20b) is degraded on this same reading.

This distinction is compatible with the argument-sensitive treatment of *all* in Križ and Spector [25] (Section 5.1): *all* removes non-maximality for its associated a rgument while allowing non-maximality in other argument positions.

Cumulative readings of numeral sentences, such as (21a), provide an importa nt motivation for the cumulative operator ∗∗ [3,12]. Their interaction with subject *all* therefore provides a relevant comparison with WR. The availability of these cu mulative readings with subject *all* is disputed, with some reports describing them a s unavailable or marginal. Examples (21a–b) are from Minor [20] (Section 3.2, exam ple 25); (21c) is from Haslinger et al. [26] (Section 1, example 1b).

(21) a. *Ten safari participants saw thirty zebras.*

b. *All the safari participants saw thirty zebras.*

c. *All the boys ate 20 sausages in total.*

On the cumulative reading of (21a), thirty specifies the total number of zebra s seen across the participants, without requiring each participant to see thirty. Mi nor reports that this reading is unavailable in (21b). However, he also acknowledg es speaker variation in comparable examples with subject *all* [20] (p. 437, n. 6). H aslinger et al. explicitly accept a cumulative reading of (21c), on which the boys'

contributions add up to twenty sausages. Their footnote 1 notes that such reading s are available in some contexts and become consistently available with expression s such as *in total*. These reports therefore support treating cumulative readings wi th subject *all* and numeral objects as disputed, whereas the WR reading of (19b) i s readily available in the peer-review context described above.

A useful comparison is provided by dependent plural readings. As discussed in Section 2.2, Winter's [6] dependency analysis allows the interpretation of a low er definite to vary with a higher distributive binder. Such dependence can coexist with *all*. Winter [6] (Section 3.5) explicitly notes that the co-distributive reading of his circle–triangle example remains available when its subject is replaced by *all cir cles*. His suggested substitution yields (22).

(22) *All circles are connected to the triangles by a dashed line.*

Adapted from Winter [6] (example 78 and the following discussion).

The relevant triangles can vary with the circle under consideration; the senten ce does not require every circle to be connected to every triangle.

Minor [20] (Section 2.3, example 15a) gives the following example:

(23) *All the boys brought their fathers along.*

On its dependent reading, each boy brings his own father; no boy needs to bring all the fathers. *All* requires participation by every boy, while the relevant fa ther varies with the boy under consideration. The grading example in (19b) exhibi ts a comparable pattern: every student participates, but the relevant partners can differ across students.

This parallel motivates a co-distributive analysis of WR. Universal requiremen ts can apply within participant-dependent partner domains without requiring ever y distinct pair to instantiate the relation. Section 4 develops this strategy using de pendent covers.

4. Formalizing reciprocity with Skolemized covers

4.1. Covers and context-sensitive distribution

The analysis retains Schwarzschild's context-sensitive approach to plural predi cation [7,27]. The distributive operator applies a predicate to contextually determin ed parts of its argument. These parts need not be atomic: distribution over larger parts permits collective and intermediate readings.

Let U be a nonempty set of atomic individuals. A cover C of U is a collectio n of nonempty subsets of U whose union is U. These subsets are its cells.

(24) $\mathrm{Cover}(C, U) \Leftrightarrow [C \subseteq (\wp(U) \setminus \{\emptyset\}) \wedge \bigcup C = U]$.

A partition requires distinct cells to be disjoint. A cover allows cells to overla p.

In the semantic formulas, each cover cell is represented by its corresponding singular or plural individual. Covers are accordingly represented as sets of such i ndividuals. The distributive operator is associated with a contextually assigned co ver and applies the predicate to every part of its argument that belongs to that c over.

(25) $[\![D_i]\!]^g = \lambda P\langle e,t\rangle.\lambda X_e.$
$\forall y \in D_e\, [(y \leq X \wedge y \in g(i)) \rightarrow P(y)]$.

Here, g(i) supplies the cover, X is the argument of distribution, and $\leq$ is the individual-part relation. The variable y ranges over atomic and plural individuals in D_e. The cover therefore determines the granularity of distribution without chan ging the universal force of D.

For illustration, consider three children a, b, and c, and a nonchild n, with U = {a, b, c, n}. For *The children danced*, the cover {{a}, {b}, {c}, {n}} requires each child to dance individually. With {{a, b, c}, {n}}, the predicate instead applies to the three-child plurality, permitting a collective dance. A cover such as {{a, b}, {c}, {n}} permits an intermediate grouping: the predicate applies to the group consisting of a and b, and separately to c.
This can describe a situation in which a and b dance together while c dances alone.

Covers also provide a representation of non-maximality, whereby plural predications tolerate contextually licensed exceptions. Following Brisson [5], this possibility arises because the cover is defined over the whole discourse universe, while D quantifies only over cells that are parts of its argument. Consider the cover {{a}, {b}, {c, n}}. The mixed cell {c, n} is not part of the children, so it falls outside the quantification in (25). The sentence consequently requires a and b to dance but leaves c unconstrained. This represents a non-maximal interpretation when the context permits c to be disregarded.

Brisson [5] calls a cover a good fit with respect to an argument if every individual member of that argument belongs to at least one cell containing only members of that argument. This ensures that nobody is excluded from the cells over which D quantifies, while allowing several members to occupy the same cell.

In the three-child example above, {{a, b}, {c}, {n}} is a good fit with respect to the children. The cells {a, b} and {c} together include *all* three children and contain no nonchildren. By contrast, {{a}, {b}, {c, n}} is ill-fitting: c occurs only in the mixed cell {c, n}, which also contains a nonchild. Since that cell is not part of the children, D does not quantify over it. The cover still covers the whole universe U, but c is left out of distribution over the children.

Following Brisson, the relevant maximality effect of subject *all* is modeled as a requirement that the associated cover be a good fit.

4.2. Dependent covers and non-maximality

Consider two teachers, Ta and Tb, who jointly teach a class of fifty students, s_1 to s_{50}. They meet with the class on separate occasions. At Ta's meeting, s_2 and s_3 are absent; at Tb's meeting, s_4 and s_5 are absent. In this context, a summary report can disregard the absence of up to three students per meeting. In this situation, sentence (26) is felicitous on a non-maximal reading.

(26) *The teachers each met their students.*

On the reading considered here, *their students* denotes the same complete class for both teachers. Every student belongs to both teachers' class, and absence from a meeting does not change this membership. The nominal argument therefore remains constant. What varies is which members may be disregarded when evaluating the meeting predicate.

A fixed lower cover selects the same student cells for both teachers. If it excludes only s_2 and s_3 from distribution, it still requires Tb to have met s_4 and s_5, who were absent from Tb's meeting. Sentence (26) is therefore not felicitous under the described scenario. Conversely, excluding only s_4 and s_5 incorrectly requires Ta to have met s_2 and s_3. Excluding all four students avoids these requirements but exceeds the contextual limit of three excluded students per meeting. It also treats the same students as exceptions for both teachers, failing to represent their different attendance patterns. A dependent cover allows the exceptions to vary with the teacher while their students continues to denote the same complete class.

An ordinary cover variable receives a cover as its value. A functional cover variable instead receives a function from individuals to covers. Applying that function to a bound individual variable supplies the cover associated with the lower distributor.

(27) a. $[\![cov_i]\!]^g = g(i)$.

b. $[\![cov_j(pro_1)]\!]^g = g(j)(g(1))$.

In (27a), g(i) is a cover of the discourse universe. In (27b), g(j) maps individuals to covers of that universe, and pro_1 supplies its individual argument. The higher distributor binds pro_1, assigning it Ta or Tb. The function g(j) remains fixed while its cover value changes with the teacher. This is the sense in which the cover is Skolemized. The semantics of D remains as in (25); the extension concerns the expression supplying its cover.

For illustration, let U consist of Ta, Tb, the fifty students, and an additional nonstudent m. The following function values represent the two exception patterns:

(28) a. $g(j)(Ta) = \{\{s_1, s_4, \ldots, s_{50}\}, \{s_2, s_3, m\}, \{Ta\}, \{Tb\}\}$.

b. $g(j)(Tb) = \{\{s_1, s_2, s_3, s_6, \ldots, s_{50}\}, \{s_4, s_5, m\}, \{Ta\}, \{Tb\}\}$.

Both values are covers of the same U. In each case, the large student cell is contained within the complete class and enters the lower distribution. The mixed cell containing m is not part of that class and is therefore excluded.
For Ta, the predicate applies to the student plurality excluding s_2 and s_3; for Tb, it applies to the plurality excluding s_4 and s_5. These larger cells yield predication of student groups. The corresponding individual-distributive interpretation can be represented by refining each large student cell into singleton cells.

The comparison with Winter's analysis concerns the location of the dependency. In Winter's [6] (Section 3.1) dependency strategy, a lower definite can identify different individuals or groups for different values of a higher binder: each soldier, for example, has their own relevant targets. The present analysis applies a similar functional architecture to the cover parameter. In the intended reading of (26), the student plurality remains the complete class, while the cover determines which parts enter predication for each teacher. The example thus illustrates how dependent covers represent participant-specific non-maximality under a fixed nominal interpretation.

4.3. Reciprocal composition and variable binding

The dependent-cover analysis retains the reciprocal denotation introduced in Section 3.1. Let A be the antecedent plurality, containing at least two atomic members. For an atomic member x of A, O(x, A) denotes the sum of all other members. The reciprocal therefore retains the complete *other*-domain; the dependent cover determines which parts of that domain enter distribution.

To make the cover expression explicit, [D cov_i] represents the operator written as D_i in (25). Replacing cov_i with $cov_j(pro_1)$ supplies a participant-dependent cover without changing the quantificational contribution of D:

(29) $[\![[D\ cov_j(pro_1)]]\!]^g = \lambda P\langle e,t\rangle.\lambda X_e.$
$\forall y \in D_e\ [(y \leq X \wedge y \in g(j)(g(1))) \rightarrow P(y)]$.

The schematic LF in (30) uses A_4 for the antecedent DP and R for the relational predicate. The range pronoun Pro_4 is anaphorically linked to the antecedent, with g(4) = A.

(30) [A_4 [[D cov_i] [1
[[the other x_1 of Pro_4]
[[D $cov_j(pro_1)$] [2 [t_1 R t_2]]]]]]]

Abstractor 1 binds the subject trace t_1, the reciprocal's contrast variable x_1, and the cover function's argument pro_1. When this binder assigns x to index 1, the reciprocal denotes O(x, A), and the lower cover expression denotes g(j)(x). Abstractor 2 binds the object trace t_2. The antecedent plurality A and the contextually as

signed function g(j) remain fixed throughout the derivation; the function's value c an vary with x.

Under the stated choice of covers, (30) is true if and only if:

(31) $\forall x \in D_{at}\,[(x \leq A \wedge x \in g(i)) \rightarrow$
$\forall y \in D_{at}\,[(y \leq O(x, A) \wedge y \in g(j)(x)) \rightarrow R(x, y)]]$.

The outer distributor applies to the antecedent members included by g(i). For each such x, the inner distributor requires R(x, y) of every other member include d by g(j)(x). Its universal force remains unchanged. Weakening arises because diff erent cover values can leave different parts of the complete *other*-domain outside distribution.

If the upper cover includes every atomic member of A and each lower cover includes every atomic member of O(x, A), (31) yields strong reciprocity.

4.4. Deriving weak reciprocity

The dependent-cover mechanism can derive a WR reading of (32), a simplifie d version of (1).

(32) *They scratched each other.*

Suppose four people, a, b, c, and d, stand in a circle. Each scratches only the next person's back: a scratches b's back, b scratches c's back, c scratches d's bac k, and d scratches a's back. Everyone scratches another person's back and has the ir own back scratched, but not every distinct pair participates.

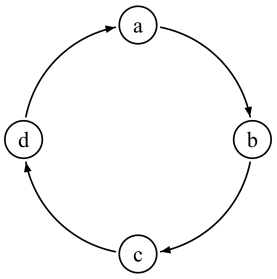


Figure 4. They scratched each other.

Let A = a ⊕ b ⊕ c ⊕ d and U = {a, b, c, d, n}, where n is outside the gro up referred to by they. The analysis can derive a WR reading in this situation us ing the following cover values:

(33) a. g(i) = {{a}, {b}, {c}, {d}, {n}}.

b. g(j)(a) = {{b}, {a, c, d, n}}.

c. g(j)(b) = {{c}, {a, b, d, n}}.

d. g(j)(c) = {{d}, {a, b, c, n}}.

e. g(j)(d) = {{a}, {b, c, d, n}}.

Each value covers the same universe U. The upper cover g(i) contains a singl eton cell for every atomic member of A, so the outer distributor applies to all fo ur participants. The lower cover depends on the participant supplied by that distr ibutor.

When the outer binder assigns a to index 1, the reciprocal denotes O(a, A), t he sum b ⊕ c ⊕ d. The lower cover is g(j)(a). Its singleton cell {b} corresponds t o a part of O(a, A), whereas its larger cell contains a and n and is therefore not

part of that argument. The lower distributor consequently requires R(a, b), without requiring R(a, c) or R(a, d).

The other function values work similarly. Distribution over O(b, A) includes c, distribution over O(c, A) includes d, and distribution over O(d, A) includes a. These are values of one fixed function g(j), evaluated at different individuals. Substituting them into (31) yields:

(34) $R(a, b) \wedge R(b, c) \wedge R(c, d) \wedge R(d, a)$.

These truth conditions entail WR. Every participant occurs as both the source and the target of a required relation. The partners in these two roles differ: a scratches b's back but has their own back scratched by d. No interaction is required to be returned by its recipient. Nor does the interpretation require all distinct pairs, although additional backscratching relations remain compatible with it.

Both instances of D retain universal force. For each participant x, the reciprocal still denotes all other members of A; g(j)(x) determines which parts of that complete plurality enter lower distribution. The circular WR interpretation is thus represented through dependent non-maximal distribution, without a dedicated cumulative operator.

This derivation illustrates the contribution of covers to the WR reading. The next subsection develops the possessive composition, and Section 4.6 examines *all*. The licensing of particular cover choices and the system's potential overgeneration are discussed in Section 4.7.

4.5. Possessive reciprocals

Section 4.4 represented backscratching as a relation between people. This subsection makes the possessive composition explicit. The proposed LF raises the entire reciprocal possessor, leaving a trace in the possessor position of *each other*'s backs. The lower distributor applies to the resulting property of possessors, within the scope of the higher distributive binder.

Let A be the plurality denoted by *they*, with g(4) = A, and let back(y) denote the back of person y. The schematic LF is:

(35) $[A_4\ [[D\ cov_i]\ [1$
$[[\text{the other } x_1 \text{ of } Pro_4]$
$[[D\ cov_j(pro_1)]\ [2$
$[t_1 \text{ scratched } [DP\ t_2 \text{ 's backs}]]]]]]]]$.

Both occurrences of D are outside the object DP. The upper D distributes over A using cov_i. Abstractor 1 binds the subject trace t_1, the reciprocal's contrast variable x_1, and the argument pro_1 of the lower cover function. For each participant x, the reciprocal therefore denotes O(x, A), and $cov_j(pro_1)$ supplies g(j)(x). The raised possessor remains within the scope of this binder, so its contrast argument is bound.

The lower D distributes over the raised reciprocal possessor. Abstractor 2 binds t_2 in the possessor position of the object DP. Unlike t_2 in Beck's reconstructed LF in (18), which occupies the reciprocal's internal range position, t_2 here replaces the entire reciprocal possessor. The two occurrences of D thus perform separate distributions without requiring separate QR of the antecedent and the range pronoun.

Composition proceeds from the lower clause. With index 1 assigned x, abstra ction over the possessor trace yields:

(36) a. λy_2.scratch(x, back(y_2)).

The lower D takes this property as its first argument and O(x, A) as its seco nd argument. With the cover g(j)(x), the result is:

b. $\forall y \in D_{at}$ [(y ≤ O(x, A) ∧ y ∈ g(j)(x))
→ scratch(x, back(y))].

The lower distribution ranges over people, while the possessive nominal map s each selected person to their back. Abstraction over index 1 then forms a prope rty of participants. The upper D applies this property to A, yielding:

(37) $\forall x \in D_{at}$ [(x ≤ A ∧ x ∈ g(i)) →
$\forall y \in D_{at}$ [(y ≤ O(x, A) ∧ y ∈ g(j)(x))
→ scratch(x, back(y))]].

Substituting the cover values in (33) gives:

(38) scratch(a, back(b)) ∧ scratch(b, back(c))
∧ scratch(c, back(d)) ∧ scratch(d, back(a)).

Every participant scratches another participant's back and has their own back scratched. Both distributors retain universal force; the dependent covers determine which pairs are required.

HLM [23] provide a specific precedent for the movement configuration. In th eir discussion of multiple reciprocals (pp. 188–189, examples 48–51), LF (51) raises an other-NP originating in possessor position while keeping it within the scope of the higher distributive binder. Their derivation involves absorption of two quanti ficational other-NPs. The present LF adopts this relative placement of the possesso r and its higher binder, while combining a group-denoting reciprocal with cover-b ased D. Extending the movement configuration to this implementation remains an assumption.

The structural difference from (18) concerns the constituent that moves. In (3 5), Pro_4 remains inside the raised reciprocal possessor, so the derivation requires no separate extraction of that pronoun from within the possessor. This avoids the particular extraction path examined in Section 3.2, but the availability of whole-po ssessor QR to the displayed clause-level position still requires independent justific ation. Nor does the comparison establish that whole-possessor movement is unava ilable to cumulative analyses. Under the stated movement assumption, (35)–(38) pr ovide a compositional derivation of possessive WR without a dedicated cumulativ e operator; they do not establish a general syntactic advantage over alternative cu mulative architectures.

4.6. "All" and the non-maximal distribution

Section 3.3 contrasted WR with the classical numeral cumulative readings that motivated analyses using ∗∗ [3,12]. The availability of these numeral readings wit h subject *all* is disputed, with some reports describing them as unavailable or mar ginal [20,26]. By contrast, WR is readily available with *all* in examples such as (19 b). The dependent-cover analysis predicts this compatibility. Subject-associated *all* r equires complete antecedent coverage, while distribution over each participant's pa rtners can remain non-maximal. This follows from distinguishing the two occurren ces of D and relating *all* to the cover associated with its nominal associate.

Brisson [28] (pp. 94–95) analyzes *all* as requiring a cover that is a good fit w ith respect to its associated plural DP. Her discussion of floating quantifiers explic itly restricts the connection between *all* and the relevant D, preventing subject-asso

ciated *all* from accessing a distributor over the object [28] (pp. 217–219). In a diffe rent framework, Križ and Spector [25] (pp. 1153, 1162–1164) likewise make *all* rem ove non-maximality at the associated argument position while allowing it elsewhe re. The present implementation applies this selective effect to the two distributive domains introduced above.

Consider the prenominal and floating occurrences of *all* in (39).

(39) a. *All the students scratched each other's backs.*

b. *The students all scratched each other's backs.*

In both sentences, *all* associates with the subject, whose denotation is A. It th erefore requires the outer cover g(i) to be a good fit with respect to A. Under th e general definition, every atomic member of A must belong to a cover cell conta ined wholly within A. In the atomic fragment used here, this reduces to (40).

(40) $\forall x \in D_{at}\, [x \leq A \rightarrow x \in g(i)]$.

Good fit restricts the admissible value of the cover; it is not an additional as sertion about scratching. Given (40), the interpretation in (37) becomes (41), where R(x, y) means that x scratched y's back.

(41) $\forall x \in D_{at}\, [x \leq A \rightarrow$
$\forall y \in D_{at}\, [(y \leq O(x, A) \wedge y \in g(j)(x)) \rightarrow R(x, y)]]$.

Every student now enters the outer distribution. The dependent covers g(j)(x), however, may still select different partners for different students: subject-associate d *all* does not require them to be good fits with respect to O(x, A). In the circula r derivation, the outer cover g(i) already includes every antecedent member and t herefore satisfies (40). Adding *all* leaves the cover values in (33) admissible and p reserves the four requirements in (38). Everyone scratches another person's back a nd has their own back scratched, without requiring every possible interaction.

The analysis also predicts that subject *all* can make a reciprocal sentence infel icitous when the situation does not support participation by every antecedent me mber.

(42) a. *The three plates are on top of each other.*

b. *# All three plates are on top of each other.*

In a finite vertical stack, physical constraints rule out a configuration in whi ch every plate is on top of another plate: there must be a bottom plate. On a no n-maximal reading of (42a), the outer cover can omit this plate from distribution. In (42b), however, outer good fit requires its inclusion, while non-vacuity requires at least one eligible partner. Together with the positive predication, these require ments would entail that the bottom plate is on top of another plate, contrary to t he possible arrangement. The degradation of (42b) on the intended distributive rea ding is therefore predicted.[2]

[2] Beck [4] (p. 128, example 205d) accepts *The books are all stacked on top of each other*, and proposes that such spatial readings require separate treatment. We make two comments. First, if we further restrict non-maximality, for instance by requiring a small antecedent group, *The three books are all stacked on top of each other* is degraded. Second, the literature on *all* acknowledges that, although *all* strongly favors maximality, it does not completely eliminate non-maximality (Križ and Spector [25], Section 5.1; Haslinger et al. [26], p. 3194, n. 39). We will not pursue this issue further here, but there is an extensive literature on the types and mechanisms of non-maximality (e.g., Malamud [29]; Križ and Spector [25]; Haslinger et al. [26]; Haslinger [30]).

The analysis makes a further prediction when *all* associates with the partner-denoting nominal. In *Each student scratched the backs of all the other students*, *all* req uires each lower cover g(j)(x) to be a good fit with respect to O(x, A). Every stu dent must therefore scratch every other student's back. This requirement can be s atisfied sequentially, with each student scratching the others in turn. *Each student scratched the backs of all the other students at exactly the same time* is predicted to be infelicitous on a reading requiring all pairwise interactions to occur simultaneousl y.

A further issue concerns actual participation. Good fit for g(i) ensures that ev ery student is evaluated, but the universal D in (41) can still be satisfied vacuousl y if a student has no eligible lower cover cell. This is possible even though every cover cell is nonempty: a cover may contain no cell wholly within O(x, A).

For the distributive reading considered here, a minimal additional requiremen t is that each participant included by the outer D have at least one eligible partn er:

(43) $\forall x \in D_{at}\,[(x \leq A \wedge x \in g(i)) \rightarrow$
$\exists y \in D_{at}\,[y \leq O(x, A) \wedge y \in g(j)(x)]]$.

The example with *all the other students* yields SR and therefore WR through c omplete outer coverage and lower good fit. WR can also be guaranteed without r equiring lower good fit. In the atomic fragment, complete outer coverage in (40) a nd non-vacuity in (43), together with the following incoming-coverage condition, a re sufficient:

(44) $\forall y \in D_{at}\,[y \leq A \rightarrow$
$\exists x \in D_{at}\,[x \leq A \wedge y \leq O(x, A) \wedge y \in g(j)(x)]]$.

Whenever (40), (43), and (44) hold, truth of (31) entails WR. Conditions (40) and (43) ensure that every antecedent member enters the outer distribution and h as at least one selected partner; (31) requires the corresponding outgoing relation. Condition (44) ensures that every member is selected as a target by another mem ber. Since that source also enters the outer distribution, (31) requires the correspo nding incoming relation. Distinctness follows from $y \leq O(x, A)$. Both relational rol es are therefore covered without requiring every distinct pair. These are sufficient conditions on a cover assignment for WR entailment.

4.7. Cover selection and overgeneration

The present analysis has an overgeneration problem. By varying covers, the s ystem can accommodate a wide range of relational patterns, including interpretati ons that speakers find unacceptable. Equation (31) specifies truth conditions relativ e to a cover assignment; it does not determine whether that assignment is context ually licensed. In particular, the existence of an assignment that makes the senten ce true in a situation does not establish that speakers can use the sentence to des cribe that situation. Formal availability within the compositional system must ther efore be distinguished from linguistic acceptability for a particular sentence and c ontext. Speakers also show clear preferences among formally available interpretatio ns. A fuller account must explain both these preferences and the exclusion of una cceptable readings. The following discussion considers predicate properties, symme try, and conversational goals as possible sources of restriction. The cited research provides independent evidence that these factors influence interpretation, but their implementation as constraints on dependent covers is a task for future research.

Research on reciprocal interpretation provides several possible sources of restr iction. Dalrymple et al.'s [2] Strongest Meaning Hypothesis selects the strongest ca

ndidate reciprocal interpretation compatible with relevant background information. Sabato and Winter [31] instead characterize interpretation domains for relational e xpressions and require maximality within those domains. Their approach makes le xical meaning, world knowledge, and context constrain the relations against which reciprocal sentences are evaluated. Neither proposal is itself a theory of depende nt cover selection, but both show why reciprocal strength cannot be left to unrest ricted contextual choice.

Symmetry is relevant, but several properties must be distinguished. A relation may be lexically symmetric, or a particular situation may contain interactions in both directions without the predicate requiring them. Moreover, pairwise symmetr y differs from WR: the backscratching circle satisfies WR without any interaction being returned by its recipient. Conversely, an asymmetric relation can form a dir ected cycle and satisfy WR. Acyclicity imposes a different restriction: on a finite p lurality, an acyclic relation cannot give every member an outgoing relation within that plurality.

These distinctions matter for *follow*. Sabato and Winter [31] (pp. 218–220) cont rast following around a Maypole, which permits a circular arrangement, with follo wing into a treehouse, which receives a path interpretation. Thus, the relevant res trictions concern the interpreted relational expression and scenario, including modi fiers, rather than an invariant classification of *follow* as incompatible with WR.

Physical possibility alone also appears insufficient. Poortman et al. [32] show that reciprocal interpretation is sensitive to the typical number of patients per age nt. Their Dutch experiment places know among predicates without a strong prefer ence for one patient and *hit* among predicates favoring one-patient situations. Thei r Maximal Typicality Hypothesis identifies a core situation that is maximal among the most typical situations for the predicate. This helps explain why sparse recip rocal configurations are more readily accepted with some predicates than with oth ers, even when denser configurations are possible. These are preferences rather th an categorical prohibitions, and the relevant tense and temporal interpretation can affect typicality.

Applied to the present proposal, these findings suggest that the availability of sparse partner selections should depend partly on predicate concepts and the im agined event structure. Knowing several people simultaneously raises different con ceptual issues from hitting several people simultaneously. Over a longer interval, however, one person can hit several others successively. A uniform restriction to one partner would therefore be inappropriate. Likewise, *know*'s tendency toward S R need not follow from lexical symmetry; it may reflect the absence of a typicalit y penalty for additional partners. Translating these preferences into restrictions on g(j) remains a task for further analysis.

Conversational goals provide another source of guidance. Malamud [29] mode ls the interpretation of plural definites through a decision problem associated with the Question Under Discussion (QUD). Križ [33] allows non-maximal uses when a situation is semantically non-false and equivalent, for current purposes, to one i n which the sentence is literally true. These approaches explain why the relevance of an exception matters, rather than merely its presence or numerical proportion. Applying this insight to covers requires a connection between discourse relevance and admissible cover values.

In the backscratching circle, spatial organization can make the person immedi ately ahead a salient partner for each participant. A report about completing the r ound may then disregard missing cross-circle interactions while treating a failure t o scratch the relevant partner as consequential. A question about whether everyon e scratched every other person's back would make those additional interactions re levant. This illustrates how a conversational goal could favor different distributive

requirements; the example does not establish a general procedure for selecting cov ers.

Crucially, contextual guidance need not identify every partner in advance. Ma lamud [29] (pp. 2–3) emphasizes that speakers need not know the exact distributi on of individuals or the identities of exceptions. A satisfactory cover analysis sho uld therefore allow a conversational standard to determine what counts as sufficie nt without always fixing one unique partner assignment. The discussion of functio nal non-maximality in Križ and Spector [25] (pp. 1171–1172) reinforces this issue: sufficient participation may vary across individuals, while their framework leaves the required extension unresolved. Dependent covers represent such variation, but the present proposal does not yet explain how context selects among, or leaves o pen, multiple suitable values.

The proposal therefore establishes a compositional means of deriving WR thr ough dependent distribution, while providing only a partial account of its context ual availability. Comparisons that hold the interaction pattern constant while varyi ng the predicate, temporal interval, or conversational goal could help determine w hich cover choices are independently supported.

5. Conclusion and open issues

This paper develops a co-distributive account of weak reciprocity through par ticipant-dependent non-maximal distribution. The reciprocal retains the complete *ot her*-domain, while a higher distributive binder supplies the individual argument of a lower cover function. Different participants can therefore be evaluated with res pect to different partner sets, while both distributive operators retain universal for ce.

The circular backscratching derivation illustrates how this mechanism yields WR. Under cover choices that give every antecedent member a nonempty partner set and include everyone as somebody else's partner, the resulting truth condition s entail participation in both relational roles. These conditions require neither pair wise symmetry nor every possible interaction. Additional interactions, including th ose satisfying SR, remain compatible with the interpretation. The derivation thus provides a compositional route to cumulative coverage without a dedicated cumul ative operator.

The possessive analysis applies the same mechanism to a property formed by abstraction over the possessor position. Its LF raises the whole reciprocal possess or, carrying the range pronoun within it, and therefore requires no separate extrac tion of that pronoun from inside the possessor. This identifies a possible structura l difference from the reconstruction of Beck's analysis examined in Section 3.2. Th e comparison remains conditional on possessor-internal extraction being blocked a nd whole-possessor QR to the proposed position being independently licensed.

The treatment of *all* distinguishes complete member participation from comple te pair coverage. Subject-associated *all* requires the outer cover to be a good fit w ith respect to the antecedent, while the participant-dependent lower covers may re main non-maximal. The outer cover in the worked WR derivation already satisfies this requirement. Adding *all* therefore preserves the partner selections and the W R reading. Together with lower non-vacuity, outer good fit ensures that every ant ecedent member participates as a source; incoming participation remains a separat e requirement. The analysis directly predicts WR's compatibility with subject all, a lthough that compatibility does not by itself exclude cumulative analyses.

The analysis still has an overgeneration problem. The current system does no t explain why speakers reject some formally available interpretations or systematic ally prefer particular readings. Predicate properties, typicality, spatial and temporal organization, and conversational goals offer possible directions for constraining co

ver selection. A general account must also explain how these factors support inco ming coverage in asymmetric WR patterns without requiring speakers to identify every partner in advance.

The contribution established here is a compositional derivation of WR within an affirmative fragment based on atomic participant relations. Further work must constrain cover selection independently, establish the syntactic licensing of the pos sessive LF, and extend the analysis to weaker reciprocal readings, collective predic ation, negation, and homogeneity.

Author Contributions: The author is responsible for the conceptualization, formal analysis, writing, and revision of this article.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable. This article reports a theoretical analysis and no new participant study.

Informed Consent Statement: Not applicable.

Data Availability Statement: No new experimental dataset was collected for this article. The analysis uses the illustrative scenarios and published sources identified in the text.

Conflicts of Interest: The author declares no conflicts of interest.

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